

(Please write your Enrolment No. immediately)

Enrolment No: | | | | | | | | | | | | | | | |

MID-TERM EXAMINATION (SET 1 / PAPER ID: 1683)
B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
B.Tech — 3rd Semester • September, 2025
Paper Code: CIC-205 • Subject: Discrete Mathematics

Time: 1½ Hours

Maximum Marks: 30

Note: Attempt Q. No. 1 which is compulsory and **any two** more questions from the remaining.

Q. No.	Question	Max. Marks	CO(s)
1 (a)	Represent the statement "If it is sunny, then I will go to the beach" using logical connectives. Write its converse, inverse, and contrapositive.	[2]	CO1
1 (b)	Prove that the statement $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$ is a tautology using a truth table.	[2]	CO1
1 (c)	Define the inclusion-exclusion principle. Also, how many numbers between 1 and 100 (inclusive) are divisible by 3 or 5 but not both?	[2]	CO4
1 (d)	Use rules of inference to demonstrate the validity of the following argument. Show how the conclusion is valid: Premises: 1. All canaries are bright yellow. 2. No large birds can sing. 3. Birds that cannot sing are a dull color. Conclusion: Canaries are small.	[2]	CO1
1 (e)	Use the Rules of Inference to show that the premises "It is not too hot and it is not too cold," and "It is either too hot or too cold," and "If it is not too cold, then the weather is great," lead to the conclusion "The weather is great."	[2]	CO1
2 (a)	Find the general solution of the Recurrence Relation: $a_n = 3a_{n-1} - 2a_{n-2} + 2^n$	[5]	CO2
2 (b)	There are 8 branches of an engineering college. What is the minimum number of students we need to select from a college to guarantee that 7 students are of the same branch?	[5]	CO2
3 (a)	Let $A = \{1, 2, 3, 4\}$. Find the power set of A, $\mathcal{P}(A)$. Define a binary relation. Give an example of a relation that is reflexive and symmetric but not transitive.	[5]	CO2
3 (b)	For each of the following functions, determine if it is injective (one-to-one), surjective (onto), and/or bijective. Justify your answer: (I) $f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(x) = x^2$ (II) $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = 2x - 3$	[5]	CO2
4 (a)	Simplify the following Boolean expression using a 4-variable Karnaugh map: $F(A, B, C, D) = \sum m(0, 2, 5, 6, 7, 8, 10, 13, 15)$ where $\sum m$ indicates the sum of minterms.	[5]	CO3
4 (b)	In how many ways can a team of 5 students be chosen from a group of 8 boys and 6 girls if the team must contain at least 2 girls?	[5]	CO4